05/02/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

***Regd. Office*:** Aakash Tower, 8,Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** **Time : 3.00 Hrs.**

Test - 8

Answer Key

1.	(4)	37.	(1)	73.	(4)	109.	(1)	145.	(3)
2.	(3)	38.	(2)	74.	(4)	110.	(2)	146	(2)
3.	(2)	39.	(1)	75.	(4)	111.	(4)	147.	(4)
4.	(3)	40.	(3)	76.	(3)	112.	(2)	148.	(2)
5.	(4)	41.	(3)	77.	(4)	113.	(4)	149.	(2)
6.	(4)	42.	(2)	78.	(4)	114.	(2)	150.	(2)
7.	(3)	43.	(4)	79.	(2)	115.	(3)	151.	(3)
8.	(1)	44.	(2)	80.	(3)	116.	(4)	152.	(2)
9.	(2)	45.	(3)	81.	(1)	117.	(4)	153.	(4)
10.	(1)	46.	(1)	82.	(2)	118.	(2)	154.	(1)
11.	(3)	47.	(3)	83.	(1)	119.	(3)	155.	(4)
12.	(4)	48.	(3)	84.	(3)	120.	(2)	156.	(4)
13.	(1)	49.	(2)	85.	(4)	121.	(1)	157.	(2)
14.	(3)	50.	(3)	86.	(1)	122.	(2)	158.	(2)
15.	(2)	51.	(2)	87.	(4)	123.	(3)	159.	(3)
16.	(1)	52.	(3)	88.	(2)	124.	(1)	160.	(4)
17.	(3)	53.	(3)	89.	(4)	125.	(1)	161.	(2)
18.	(1)	54.	(3)	90.	(2)	126.	(4)	162.	(4)
19.	(4)	55.	(3)	91.	(4)	127.	(2)	163.	(1)
20.	(1)	56.	(1)	92.	(3)	128.	(4)	164.	(2)
21.	(4)	57.	(3)	93.	(2)	129.	(2)	165.	(3)
22.	(1)	58.	(4)	94.	(4)	130.	(4)	166.	(4)
23.	(1)	59.	(3)	95.	(2)	131.	(3)	167.	(4)
24.	(3)	60.	(4)	96.	(3)	132.	(2)	168.	(2)
25.	(3)	61.	(2)	97.	(3)	133.	(3)	169.	(3)
26.	(3)	62.	(2)	98.	(2)	134.	(4)	170.	(2)
27.	(3)	63.	(2)	99.	(2)	135.	(3)	171.	(3)
28.	(2)	64.	(1)	100.	(2)	136.	(2)	172.	(4)
29.	(1)	65.	(3)	101.	(4)	137.	(4)	173.	(3)
30.	(3)	66.	(2)	102.	(4)	138.	(3)	174.	(4)
31.	(3)	67.	(3)	103.	(3)	139.	(2)	175.	(3)
32.	(4)	68.	(2)	104.	(2)	140.	(4)	176.	(2)
33.	(2)	69.	(3)	105.	(4)	141.	(4)	177.	(4)
34.	(1)	70.	(4)	106.	(2)	142.	(3)	178.	(4)
35.	(4)	71.	(3)	107.	(2)	143.	(1)	179.	(4)
36.	(2)	72.	(2)	108.	(1)	144.	(2)	180.	(1)

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Test - 8

Hints and Solutions

PHYSICS

1. Answer (4)

$$E = \frac{B\omega}{2}\left(r_f^2 - r_i^2\right)$$

$$= \frac{B\omega}{2}\left(\left(\frac{\ell}{2}\right)^2 - \left(\frac{\ell}{2}\right)^2\right)$$

= 0

2. Answer (3)

$$\phi = Li \Rightarrow L = \frac{\phi}{i} = \frac{4}{2} = 2\text{ H}$$

3. Answer (2)

$\phi = 4t - 2t^2 = 0$

$\Rightarrow \ 4t = 2t^2$

$\Rightarrow \ t = 2$ s

$$E = \left|\frac{d\phi}{dt}\right| = \left|\frac{d(4t - 2t^2)}{dt}\right| = |\,4 - 4t\,|$$

$= |4 - 4 \times 2| = 4$ V (at $t = 2$ s)

4. Answer (3)

The line joining places with equal angles of dip are called isoclinic lines.

5. Answer (4)

The phenomenon of perfect diamagnetism in superconductors is called Meissner effect.

6. Answer (4)

P_{out} = 50% of P_{in}

$$2000 \times i = \frac{50}{100} \times 10 \times 100$$

$$i = \frac{50}{100} \times \frac{10 \times 100}{2000} = 0.25\text{ A}$$

7. Answer (3)

$$z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{4^2 + (2-2)^2} = 4\ \Omega$$

8. Answer (1)

$$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{20}{\sqrt{2}}$$

This value is same for both voltages

$\therefore$ Ratio = 1 : 1

9. Answer (2)

$$I_{rms} = \sqrt{I_{dc}^2 + \frac{I_{ac}^2}{2}}$$

$$= \sqrt{4^2 + \frac{2^2}{2}}$$

$$= \sqrt{4^2 + 2}$$

$$= \sqrt{16 + 2} = 3\sqrt{2}\text{ A}$$

10. Answer (1)

$$e = \frac{L\Delta i}{\Delta t} = 10 \times 10^{-3} \times \frac{(2-1)}{2} = 5\text{ mV}$$

11. Answer (3)

$i_{\text{effective}} = i_{\text{peak}} / \sqrt{2}$

$= \frac{523}{1.41} = 370$ V

12. Answer (4)

Displacement current, $I_d = A\varepsilon_0 \frac{dE}{dt} = \frac{A\varepsilon_0}{d}\frac{dV}{dt}$

So, the potential difference has to change with time.

13. Answer (1)

$I_d = \varepsilon_0 A\frac{dE}{dt} = \varepsilon_0(\pi R^2)\frac{dE}{dt} = i$

$I_d' = \varepsilon_0\left(\pi R^2 - \frac{\pi R^2}{4}\right)\frac{dE}{dt}$

$= \frac{3}{4}\varepsilon_0 \pi R^2 \frac{dE}{dt}$

$I_d' = \frac{3}{4}i$

14. Answer (3)

Conduction current is the electric current which exists in a conductor when the electrons flow in the conductor.

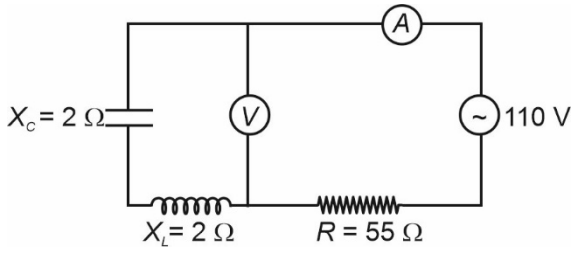
15. Answer (2)

Current through capacitor,

$I = \frac{E}{X_C} = \frac{E}{\frac{1}{\omega C}} = \omega CE = 2\pi\nu\, CE$ or $I \propto \nu$

$\therefore$ Decrease in frequency ν of ac source decreases the conduction current. As displacement current is equal to conduction current, decrease in ν, decreases displacement current in circuit.

16. Answer (1)



$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$= \sqrt{(55)^2} = 55\ \Omega$

$V = IZ$

$\frac{110}{55} = I \Rightarrow I = 2$ A

17. Answer (3)

$f_1 = 500 \times 10^3$ Hz

$L' \to 2L, \quad C' \to \frac{1}{8}C$

$f = \frac{1}{2\pi\sqrt{CL}} \Rightarrow f \propto \frac{1}{\sqrt{CL}}$

$\frac{f_1}{f_2} = \frac{\sqrt{L_2C_2}}{\sqrt{L_1C_1}} = \sqrt{\frac{(2L)\left(\frac{1}{8}C\right)}{LC}}$

$\frac{f_1}{f_2} = \frac{1}{2} \Rightarrow f_2 = 1000$ kHz

18. Answer (1)

Power factor

$\cos\phi = \frac{R}{\sqrt{R^2 + X_L^2}} = \frac{6}{\sqrt{6^2 + 6^2}} = \frac{1}{\sqrt{2}}$

$\Rightarrow$ Phase angle = 45° = ϕ

19. Answer (4)

$U_{\text{Inductor}} = U_{\text{Total}} - U_{\text{Capacitor}}$

$U_{\text{Inductor}} = \frac{Q^2}{2C} - \frac{\left(\frac{Q}{4}\right)^2}{2C} = \frac{15U}{16}$

20. Answer (1)

Here, $C = 30\ \mu\text{F} = 30 \times 10^{-6}$ F,

$L = 27$ mH $= 27 \times 10^{-3}$ H

$\therefore \quad \omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{27 \times 10^{-3} \times 30 \times 10^{-6}}}$

$= \frac{1}{\sqrt{81 \times 10^{-8}}}$

$= \frac{10^4}{9} = 1.1 \times 10^3$ rad s^{-1}

21. Answer (4)

$\omega^2 = 64$

$\omega = 8$

$\Rightarrow \quad 2\pi f = 8 \Rightarrow f = \frac{4}{\pi}$

22. Answer (1)

Here, $C = 40\ \mu F = 40 \times 10^{-6}$ F

$L = 16$ mH $= 16 \times 10^{-3}$ H

$\omega = 1/\sqrt{LC} = 1/\sqrt{(16\times10^{-3}\times40\times10^{-6})}$

$= (1/8) \times 10^4 = 1250$ rad/s

$T = \frac{2\pi}{\omega} \simeq 5$ ms

23. Answer (1)

$\varepsilon = Bvl$

$= 4\times10^{-4}\times360\times\frac{5}{18}\times25 = 1$ V

24. Answer (3)

$\phi = Mi$

$\frac{d\phi}{dt} = \frac{Mdi}{dt}$

$1000 = \frac{M(2)}{0.01}$

$M = \frac{1}{2}(10) = 5.00$ H

25. Answer (3)

$Z = \sqrt{R^2 + X_L^2} = \sqrt{4^2 + 3^2} = 5\ \Omega$

$i_{peak} = \frac{V_{peak}}{Z} = \frac{10}{5} = 2$ A

26. Answer (3)

$Q = \frac{\Delta\phi}{R} = \frac{8-4}{10} = 0.4$ C

27. Answer (3)

At resonance the phase difference between potential and current is zero.

28. Answer (2)

$\eta = \frac{P_{out}}{P_{in}}\times100 = \frac{100\times100}{3\times200} = \frac{1}{6}\times100\%$

$= \frac{50}{3}\% = 16.6\%$

29. Answer (1)

$I_E = I_B \Rightarrow \frac{I_E}{I_B} = 1$

30. Answer (3)

Total power = Solar constant × area

$= 10^4 \times (20 \times 10) = 2 \times 10^6$ W

31. Answer (3)

$F = \frac{2IA}{c} = \frac{2\times30\times10\times10^{-4}}{3\times10^8}$

$= \frac{600\times10^{-4}}{3\times10^8}$

$= 200\times10^{-12} = 2\times10^{-10}$ N

32. Answer (4)

$I = I_0 \sin(\omega t)$

$= 10\sqrt{2}\sin\left(2\pi\times\frac{1}{400}\times50\right)$

$= 10\sqrt{2}\sin\left(\frac{\pi}{4}\right) = 10$ A

33. Answer (2)

$X_C = \frac{1}{\omega C}$

34. Answer (1)

$E = (\vec{v}\times\vec{B})\cdot\vec{\ell}$

$= \left\{\left(4\hat{i}+\hat{j}\right)\times\left(8\hat{i}+2\hat{j}\right)\right\}\cdot\vec{\ell} = 0$

35. Answer (4)

Susceptibility of a diamagnetic substance is independent of temperature.

36. Answer (2)

Beyond Curie temperature, a ferromagnetic substance behaves as a paramagnetic substance.

37. Answer (1)

$I = \chi H$

$= (6 \times 10^{-4})(8 \times 10^{-4})$

$I = 48 \times 10^{-8}$ A/m

38. Answer (2)

$I_{rms} = \frac{I_0}{\sqrt{2}}$

$I_{mean} = \frac{2I_0}{\pi}$

$\frac{I_{mean}}{I_{rms}} = \frac{2\sqrt{2}}{\pi}$

39. Answer (1)

$\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{(X_L - X_C)^2 + R^2}}$

$\frac{R}{R} = 1$ (as $X_L = X_C$)

40. Answer (3)

$P_{dc} = i^2R = 4^2R$

$P_{ac} = i_{rms}^2 R = \left(\frac{4}{\sqrt{2}}\right)^2 R$

$\frac{P_{dc}}{P_{ac}} = 2$

41. Answer (3)

$i = \frac{V}{R} = \frac{100}{10} = 10$ A

$U = \frac{1}{2}Li^2 = \frac{1}{2} \times 5 \times 10^{-3} \times 10^2$

$= 250 \times 10^{-3}$ J $= 0.25$ J

42. Answer (2)

$L = \frac{\mu_0 N^2 A}{l}$

$L \propto \frac{1}{l}$

43. Answer (4)

Given: $E_{rms} = 18$ V/m

$B_{rms} = \frac{E_{rms}}{C} = \frac{18}{3 \times 10^8} = 6 \times 10^{-8}$

$B_0 = B_{rms}\sqrt{2} = 6\sqrt{2} \times 10^{-8} = 8.5 \times 10^{-8}$ T

44. Answer (2)

Accelerated charges produce electromagnetic waves.

45. Answer (3)

$e = B\left(\frac{dA}{dt}\right)$

$= B\frac{dA}{dt}(x^2)$

$= B(2x)\left(\frac{dx}{dt}\right)$

$= 2Ba\alpha$

CHEMISTRY

46. Answer (1)

Magnetic moment

$= \sqrt{n(n+2)}$ BM

$= \sqrt{1 \times 3} = \sqrt{3} = 1.732$ BM

47. Answer (3)

Fe : [Ar] $3d^6\ 4s^2$

Fe^{3+} : [Ar] $3d^5$,

Yellow colour ion of having five unpaired electron.

48. Answer (3)

With increase in oxidation state of centre atom in a compound, acidic strength increases.

49. Answer (2)

Interstitial compounds retain metallic conductivity.

50. Answer (3)

$4FeCr_2O_4 + 8K_2CO_3 + 7O_2 \rightarrow 8K_2CrO_4 + 2Fe_2O_3 + 8CO_2$

51. Answer (2)

Elements of 4*d* and 5*d* series (of same group) are having almost similar size.

Mo and W are having almost similar size.

52. Answer (3)

In 3*d*-series elements, Zn has least enthalpy of atomisation.

53. Answer (3)

$2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

54. Answer (3)

$Cr_2O_7^{2-} + H_2\overset{-2}{S} \rightarrow Cr^{3+} + S^0$

Mole of $Cr_2O_7^{2-} \times 6$ = Mol of $H_2S \times 2$

Mole of $Cr_2O_7^{2-} \times 3$ = Mol of H_2S

Mole of $Cr_2O_7^{2-} \times 3 = 0.3$

Mole of $Cr_2O_7^{2-} = 0.1$

55. Answer (3)

Yb (II) is a reductant.

56. Answer (1)

Electronic configuration of

Sm(Samarium, 62) is : $[Xe]4f^6\ 6s^2$

57. Answer (3)

On moving left to right across Lanthanoids ionic size of trivalent ions regularly decreases.

58. Answer (4)

Actinium (Ac) only shows +3 oxidation state.

59. Answer (3)

5*f* electrons of actinoids are more effectively shielded from the nuclear charge than 4*f*-electrons of the corresponding lanthanoids.

60. Answer (4)

Order of melting point : W > Ta > Ir > Au

61. Answer (2)

en (Ethylenediamine) is a neutral bidented ligand.

62. Answer (2)

Order of field strength :

$OH^- < C_2O_4^{2-} < H_2O < CN^-$

63. Answer (2)

In low spin complex, pairing of electrons occur, so correct configuration will be

$t_{2g}^5\ e_g^0$.

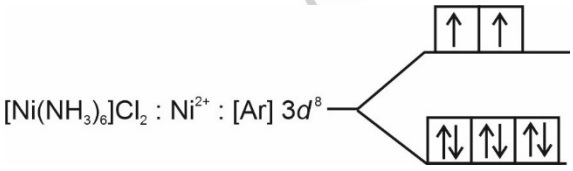
64. Answer (1)

One bidented ligand like en and ox forms one ring.

65. Answer (3)

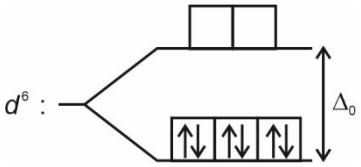
$[Co(NH_3)_2(H_2O)_2Br_2]Cl$ can show optical isomerism.

66. Answer (2)



So, number of unpaired electrons = 2.

67. Answer (3)



CFSE = [–0.4 × 6] Δ_0 + 3P

= – 2.4 Δ_0 + 3P

68. Answer (2)

Metal having most positive charge, will causes least back donation and has largest C–O bond order.

69. Answer (3)

$Na_3[FeCl_6]$ complex has Fe^{3+} ion which has 5 unpaired electrons, so it will exhibit maximum paramagnetic behaviour.

70. Answer (4)

- en and gly both are bidented ligands.
- Primary valency = Oxidation state = 2

Secondary valency = Co-ordination number = 6

71. Answer (3)

Due to presence of vacant orbitals in CO molecule, it can accept e^- density from metal/metal ion.

72. Answer (2)

Type of ligands present in homoleptic complex are same.

73. Answer (4)

EAN = Atomic number – Oxidation state
+ Number of electrons obtained from ligands
= 26 – 1 + 12 = 37

74. Answer (4)

$CoCl_3.6NH_3 \Rightarrow [Co(NH_3)_6]Cl_3$

$\Rightarrow [Co(NH_3)_6]^{3+} + 3Cl^-$

75. Answer (4)

Complex containing chelating ligands are generally more stable than complex containing monodentate ligands.

76. Answer (3)

In polar aprotic medium, anions will not solvate and nucleophilicity in this case depends on only charge density.

77. Answer (4)

Electron withdrawing group(s) present on ortho and para position will increase rate of nucleophilic substitution.

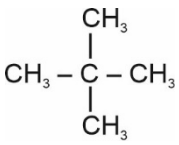
78. Answer (4)

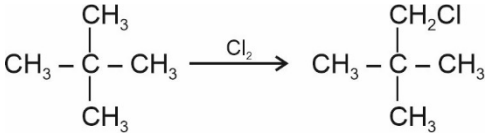
Ortho substituted biphenyl compounds having no plane of symmetry is optically active.

79. Answer (2)

3°-carbocation (of 9α-H) is more stable than of benzylic carbocation.

80. Answer (3)

Compound  on monochlorination with Cl_2 in the presence of UV light gives only one product.



81. Answer (1)

Fluoro derivatives of hydrocarbons generally having less density than water.

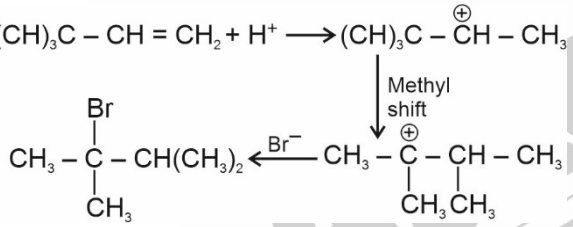
82. Answer (2)

E1 reaction involves carbocation as intermediate, for which correct order of stability is 3° > 2° > 1°.

83. Answer (1)

In Fittig reaction, haloarenes in the presence of dry ether gives diarenes.

84. Answer (3)



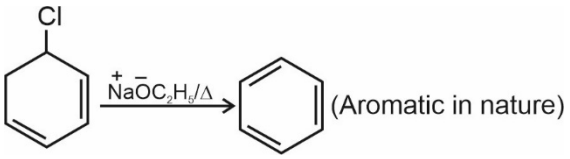
85. Answer (4)

Compounds which are optically active or having chiral centre give racemisation on hydrolysis.

86. Answer (1)

Due to presence of less steric crowding in CH_3Cl, it gives fastest rate of S_N2.

87. Answer (4)

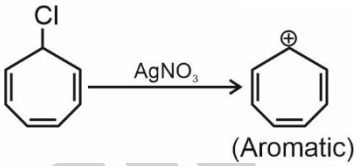


88. Answer (2)

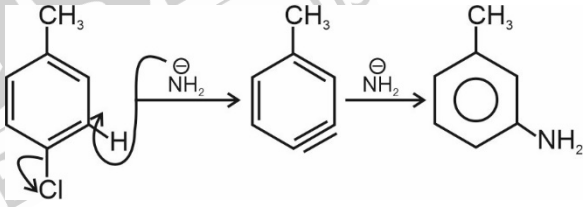
Correct order of dipole moment is

$CH_3Cl > CH_3F > CH_3Br > CH_3I$

89. Answer (4)



90. Answer (2)



BOTANY

91. Answer (4)

Nitrogenous base + Sugar = Nucleoside

92. Answer (3)

E. coli has 4.6×10^6 bp of DNA.

93. Answer (2)

Acidic substance present in nucleus first identified by Friedrich Miescher in 1869 was DNA and he named it 'Nuclein'

94. Answer (4)

5-methyl uracil (thymine) is found in DNA only.

95. Answer (2)

The unequivocal proof that DNA is the genetic material came from the experiments of Alfred Hershey and Martha Chase (1952)

96. Answer (3)

RNA performs the dynamic functions of messenger and adapter. It has ribose sugar with 2'OH group and has uracil instead of thymine.

97. Answer (3)

It should provide the scope for slow changes (mutation) that are required for evolution

98. Answer (2)

DNA can be synthesised from RNA by reverse transcriptase enzyme.

99. Answer (2)

Thymine is absent in RNA and instead of thymine it has uracil.

100. Answer (2)

DNA replicates semi conservatively in both *E.coli* and humans

101. Answer (4)

Average rate of DNA polymerization in prokaryotes is approximately 2000 bp per second

102. Answer (4)

DNA polymerase I is DNA-dependent DNA polymerase enzyme.

103. Answer (3)

Helicase unwind the DNA double helix during DNA replication.

104. Answer (2)

23 S rRNA also known as peptidyl transferase catalyzes the peptide bond formation in prokaryotes.

105. Answer (4)

RNA polymerase III is responsible for the synthesis of 5 S rRNA, tRNA and snRNA in eukaryotes.

106. Answer (2)

Rho factor is responsible for termination of transcription in prokaryotes.

107. Answer (2)

Severo Ochoa enzyme (polynucleotide phosphorylase) is helpful in polymerising RNA with defined sequences in a template independent manner (enzymatic synthesis of RNA)

108. Answer (1)

$5' \rightarrow 3'$ strand of DNA is the coding strand of DNA.

109. Answer (1)

In tailing, adenylate residues (200-300) are added at 3'-end in a template independent manner by Poly A polymerase enzyme.

110. Answer (2)

Introns are generally present in eukaryotes

111. Answer (4)

AUG is a start codon in both prokaryotes and eukaryotes.

112. Answer (2)

There are two sites in the large subunit of ribosome, for subsequent amino acids to bind to and thus, be close enough to each other for the formation of a peptide bond.

113. Answer (4)

First three triplet codons are used to synthesize three amino acids and UAG is stop codon.

114. Answer (2)

The repressor protein binds to the operator region of the operon and prevents RNA polymerase from transcribing the operon.

115. Answer (3)

Lac z encodes for beta galactosidase enzyme.

116. Answer (4)

In actual structure the tRNA is a compact molecule and looks like inverted L.

117. Answer (4)

In prokaryotes, control of rate of transcriptional initiation is the predominant site for control of gene expression.

118. Answer (2)

Euchromatin is loosely packed, stains light and transcriptionally active.

119. Answer (3)

HGP was completed in 2003

120. Answer (2)

DNA fingerprinting involved southern blot hybridisation using radio labelled VNTR as a probe. VNTR is called mini-satellite.

121. Answer (1)

Chromosome 1 has most genes (2968), and the Y has the fewest (231).

122. Answer (2)

Himgiri is a variety of wheat having the resistance to leaf and stripe rust.

123. Answer (3)

Changes in the base sequence within genes through use of chemicals or radiations and selecting and using the plants that have the desirable characters as a source of breeding, is called mutational breeding.

124. Answer (1)

Pusa swarnim is a variety of *Brassica* which is resitant to white rust.

125. Answer (1)

Parbhani kranti variety of *A. esculentus* is resistant to yellow mosaic virus.

126. Answer (4)

Pusa Komal is a variety of cowpea.

127. Answer (2)

Atlas-66 is a variety of wheat having high protein content.

128. Answer (4)

International Rice Research Institute is located at Philippines.

129. Answer (2)

Taichung-1 is a semi-dwarf rice variety developed in Taiwan.

130. Answer (4)

Tropical canes grown in south India, *Saccharum officinarum* had thicker stems and higher sugar content but did not grow well in north India.

131. Answer (3)

In mung bean, resistance to yellow mosaic virus and powdery mildew were induced by mutations.

132. Answer (2)

Pomato was developed through the somatic hybridization.

133. Answer (3)

Plant cells are totipotent, and it is the basis of tissue culture.

134. Answer (4)

Meristem culture is used to obtain virus free plants.

135. Answer (3)

Biofortification can improve the nutritional quality of food and take care of silent hunger.

ZOOLOGY

136. Answer (2)

In their simulation experiment, they proved the formation of organic molecules from inorganic constituents.

137. Answer (4)

Early formed cells were chemoheterotrophs, thermophilic, prokaryotic and anaerobic in nature.

138. Answer (3)

They used CH_4 : NH_3 and H_2 in ratio 2 : 1 : 2.

139. Answer (2)

RNA has both self replicating and catalytic properties.

140. Answer (4)

Coacervates do not have lipid membrane and cannot reproduce; theory of biogenesis confirms that life arises from pre-existing life. Present day atmosphere is oxidising type.

141. Answer (4)

It is a type of directional selection.

142. Answer (3)

Auricular muscles in humans are vestigial

143. Answer (1)

Population is the smallest unit where evolutionary changes occur while natural selection acts upon individual.

144. Answer (2)

Formation of many species from a single species is speciation.

145. Answer (3)

These are the examples showing marsupial radiations in Australia.

146 Answer (2)

Cycads flourish in Jurassic period of Mesozoic era.

147. Answer (4)

Flippers of penguin and dolphin are analogous organs.

148. Answer (2)

Extant means still in existence.

149. Answer (2)

Silurian period is a part of Palaeozoic era.

150. Answer (2)
Adaptive radiation leads to divergent evolution.

151. Answer (3)
Crocodiles, dinosaurs and birds originate from thecodonts.

152. Answer (2)
Darwin finches provide an evidence related to the distribution of species.

153. Answer (4)
The original Darwin finches were seed eaters.

154. Answer (1)
Darwinian variations were small and directional and mutations are random and directionless.

155. Answer (4)
Nature support those features which are highly adaptive.

156. Answer (4)
Natural selection works and more organism reach upto adulthood and contribute more to the gene pool.

157. Answer (2)
During his voyage, he went on H.M.S Beagle.

158. Answer (2)
Genetic drift is a per chance change in gene frequency which can be due to founder's effect or bottle neck effect.

159. Answer (3)
According to Darwin, fitness level means who leave many living descendants in a gene pool.

160. Answer (4)
Only germplasm DNA is inheritable.

161. Answer (2)
In directional/Progressive selection, more individuals acquire value other than mean character value.

162. Answer (4)
Recapitulation theory was given by Ernst Haeckel.

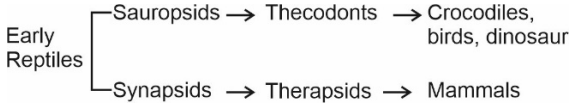
163. Answer (1)
Use and disuse of organs are related with Lamarckism.

164. Answer (2)
According to Hardy Weinberg equilibrium,
$(p + q)^2 = p^2 + 2pq + q^2 = 1$

165. Answer (3)
Darwin finches are examples of founders effect.

166. Answer (4)
Oparin gave theory of origin of life.

167. Answer (4)



168. Answer (2)
Around 10,000 years back agriculture came into existence.

169. Answer (3)
Homo habilis – 650-800 CC ;
Homo erectus – 900 CC
Australopithecus – 500 CC ;
Homo neanderthalensis – 1400 CC

170. Answer (2)
Prehistoric cave art can be seen in Bhimbetka rock shelter in Raisen district, Madhya Pradesh.

171. Answer (3)
In adult chimpanzee, skull grows at faster rate than body.

172. Answer (4)
Homo neanderthalensis cover their body with animal hides and buried their dead.

173. Answer (3)
Eyes of *Octopus* and fish are analogous organs show convergent evolution.

174. Answer (4)
Option 1, 2, 3 are related with the adaptations.

175. Answer (3)
Absence of random mating / selective mating disturbs Hardy Weinberg equilibrium.

176. Answer (2)
Ice age is between 75,000 – 10,000 years ago.

177. Answer (4)
Homo erectus were probably meat eaters.

178. Answer (4)
About 65 mya, dinosaurs disappeared from earth.

179. Answer (4)
Mole = marsupial mole; mouse – Marsupial mouse, Lemur = spotted cuscus.

180. Answer (1)
Marsupials survived in Australia due to lack of competition from other mammals.

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